QoR Surrogate Modeling for Early HLS-to-ASIC Decision Making

Berk Yonder - 10994918
Politecnico di Milano - Advanced Computer Architectures
Instructor: Prof. Christian Pilato
June 29, 2026

Abstract

This report presents an HLS to ASIC quality-of-results (QoR) surrogate modeling workflow for early design-space decision making. The pipeline extracts static features from PolyBench/C source files [1], Bambu/HLS metrics, and Yosys synthesis statistics before final physical design, and uses these features to predict OpenROAD-reported ASIC QoR labels. The main prediction targets are final ASIC area and critical-path delay, while total power is treated as an exploratory target because its observed distribution is skewed and sensitive to clock constraints. Under benchmark-grouped validation, the best direct-target models achieved 15.34% normalized mean absolute error (NMAE) for final area and 18.71% NMAE for critical-path delay. Power prediction was less reliable, with 37.70% NMAE. The report also includes a physical-effect analysis of routed wirelength, route-guide indicators, and route-DRC indicators to study how backend routing behavior relates to final QoR. The results show that early source-level, HLS, and synthesis features contain meaningful predictive signal for final ASIC area and timing, while the small OpenROAD-labeled subset limits the statistical confidence and generality of the predictor.

Keywords: High-level synthesis, ASIC design, OpenROAD, Yosys, QoR prediction, Surrogate modeling, Regression, Design-space exploration, Machine learning.

Repository: github.com/berkyonder/qor-surrogate-modeling

1 Introduction

High-level synthesis (HLS) enables hardware design points to be generated from high-level source kernels, such as C/C++ programs, by changing constraints such as clock period, memory allocation policy, arithmetic resource allocation, and optimization level. However, the final ASIC quality of each design point is only known after downstream physical implementation. Running a complete OpenROAD backend flow for every HLS configuration is expensive, so evaluating all design-space points only through place and route is not practical.

The objective of this project is to study whether final ASIC QoR can be predicted from information available before backend physical implementation, which is represented in this project by the final OpenROAD flow. The problem is formulated as a supervised regression task. Input features are extracted from static source-code analysis, Bambu/HLS reports, and Yosys synthesis reports. Target labels are parsed from OpenROAD backend reports. The main targets are OpenROAD-reported final ASIC area and critical-path delay. Total power is included as an exploratory target because the observed power distribution is strongly affected by clock conditions.

The intended use case is early HLS to ASIC decision making. A surrogate model with acceptable prediction error can rank, filter, or prioritize design configurations before executing the expensive backend flow. The method does not replace physical design; instead, it can reduce the number of backend candidates that need complete implementation by identifying promising, poor, or uncertain configurations earlier in the design-space exploration process.

This report also analyzes physical effects such as timing and electrical violations, approximate routed wirelength, route-guide indicators, and route-DRC indicators. These descriptors are extracted from OpenROAD outputs and are therefore not used as early-prediction inputs. They are used only to interpret how final physical-design behavior relates to final QoR.

The main contributions of the project are:

- An end-to-end report-parsing and dataset-construction pipeline from HLS to OpenROAD;

- A benchmark-grouped regression evaluation for final area, delay, and power;
- A feature-group analysis comparing the contribution of static, HLS, synthesis, and derived-ratio information;
- A post-routing physical-effect analysis connecting routing descriptors to final QoR.

2 Workflow and Experimental Setup

The project was implemented as an end-to-end Python pipeline that converts raw HLS, synthesis, and backend reports into structured datasets for regression modeling. The overall flow is:

PolyBench/C source → Static features → Bambu/HLS metrics
→ Yosys statistics → OpenROAD QoR labels → Regression models

The experimental setup used Ubuntu virtual machines running in Oracle VirtualBox on a Windows host. The Bambu and early feature-extraction environment was configured with 2 virtual processors, 4096 MB RAM, and a 40 GB virtual disk. The OpenROAD backend environment was configured with 4 virtual processors, 8192 MB RAM, and a 60 GB virtual disk. This separation was useful because OpenROAD-flow-scripts required a heavier Linux environment and more computational resources than early HLS parsing or Python-based modeling.

Table 1: Virtualized experimental environments

Environment	RAM	CPUs
Bambu / early pipeline VM	4096 MB	2
OpenROAD backend VM	8192 MB	4

In practice, the dataset was created through several script-based stages rather than taken from a pre-existing benchmark table. The early-stage scripts enumerated combinations of benchmark, dataset size, Bambu clock period, memory allocation policy, DSP coefficient, and optimization level. Bambu/HLS logs were then parsed to recover both the configuration information and the reported HLS metrics. Static source-code features were extracted directly from the PolyBench/C files, and Yosys reports were parsed for the subset of designs with available synthesis results. Since full OpenROAD backend implementation is more expensive, only a smaller benchmark-diverse subset was prepared and executed through OpenROAD-flow-scripts.

Within these Ubuntu guest environments, Python was used for data processing, feature extraction, model training, validation, and plotting. The main libraries were `pandas`, `numpy`, `scikit-learn`, and `matplotlib`. Bambu was used for HLS report generation, Yosys for RTL synthesis statistics, and OpenROAD-flow-scripts with the Nangate45 platform for backend ASIC reports.

Raw Bambu logs are parsed by `parse_bambu_log.py`. This script extracts scheduling, resource, and early area-related metrics such as control steps, estimated frequency, state count, flip-flops, register estimates, mux area, total estimated area, and performance conflicts. The same parser reconstructs the design-space configuration from filenames, including benchmark, dataset size, clock period, memory policy, DSP coefficient, and optimization level.

Static source-code features are extracted by `extract_static_features.py`. This script analyzes PolyBench/C files using lightweight pattern-based parsing, based on source-text patterns such as loop keywords, conditional keywords, array-index expressions, arithmetic operators, assignment operators, function-call patterns, and declaration patterns. From these patterns, it records structural properties such as source lines, loop counts, approximate loop depth, conditional statements, array accesses, arithmetic operations, assignments, function calls, and declarations. These features are available before HLS and therefore represent the earliest source-level information in the pipeline.

Synthesis-level features are extracted from Yosys reports by `parse_yosys_reports.py`. Since the HLS-generated Verilog can be hierarchical, the parser records both selected implementation-module statistics and aggregate module statistics. Extracted values include wire counts, wire-bit counts, public wire counts, memory counts, process counts, cell counts, and module counts. These features are still pre-OpenROAD because they are obtained before final placement and routing.

OpenROAD reports are parsed by `parse_openroad_reports.py`. This step extracts the final regression labels: chip area, critical-path delay, and total power. It also extracts timing and electrical closure quantities such as WNS, TNS, worst slack, minimum clock period, maximum frequency, setup violations, hold violations, slew violations, and capacitance violations.

The datasets are merged by `preprocess_datas` et.py. This step aligns configuration keys, combines Bambu, static-code, Yosys, and OpenROAD tables, removes unused identifier or status columns, one-hot encodes categorical variables, and writes the processed modeling datasets. Regression experiments are then run using baseline models, feature-engineered models, regularized direct-target models, exploratory target variants, and feature-group comparisons.

Finally, `parse_openroad_physical_reports.py` extracts approximate routed wirelength from final DEF coordinate sequences, route-guide indicators, and conservative route-DRC indicators. `analyze_openroad_physical_correlations.py` correlates these physical descriptors with final OpenROAD QoR targets.

3 Dataset Construction

The early design space was generated from PolyBench/C kernels, following the PolyBench benchmark setting used in Bambu QoR studies [1]. The grid combines benchmark, dataset size, Bambu clock period, memory allocation policy, DSP allocation coefficient, and optimization level. This produced 864 early HLS configurations, which provide the broadest view of the design space because they are cheaper to generate than full backend implementations.

The dataset becomes smaller as more expensive labels are added. After report availability and parsing checks, 79 configurations had usable Yosys synthesis statistics and 18 configurations had complete OpenROAD backend labels. The OpenROAD-labeled subset includes 17 PolyBench benchmarks in the main 10 ns backend subset, plus one additional aggressive-clock run used to explore clock sensitivity. This staged construction reflects the increasing cost and complexity of the flow from HLS to synthesis and then to full physical implementation.

Before modeling, the processed datasets were checked using data-quality reports. These checks included missing-value summaries, duplicate detection, constant-column detection, outlier summaries, group-level summaries by benchmark and configuration variables, and correlation reports with the main QoR targets. These checks were used to verify that the merged dataset was suitable for regression modeling and to identify potentially uninformative or problematic features.

Table 2: Dataset progression to OpenROAD labels

Stage	Rows	Role
Full HLS	864	Early design-space configurations
Yosys subset	79	Adds synthesis structure
OpenROAD subset	18	Final ASIC QoR labels
Main backend subset	17	Benchmark-diverse 10 ns runs
Aggressive-clock run	1	Clock-sensitivity probe

This progression is central to the interpretation of the results. The 864-row early dataset shows that the pipeline can cover a broad HLS design space, but the supervised OpenROAD regression task is limited by the 18 backend-labeled rows. Consequently, the numerical results should be read as a feasibility study rather than as evidence of a mature production predictor.

4 Feature and Target Definition

The surrogate-model inputs are intentionally limited to information available before final OpenROAD routing. Table 3 summarizes the main feature groups used by the modeling scripts.

Table 3: Pre-OpenROAD feature groups

Group	Examples
Config.	Clock, dataset size, memory policy, DSP coefficient, optimization level
Static code	Lines, loops, loop depth, branches, array accesses, operations
Bambu/HLS	Control steps, states, frequency, area estimate, registers, mux area
Yosys	Wires, wire bits, cells, memories, processes, modules
Ratios	Compute/memory, control density, wires per cell, area per step

The model outputs are continuous final QoR quantities, so the task is regression. Table 4 lists the primary OpenROAD targets.

Table 4: Final OpenROAD QoR targets

Target	Role
`openroad_synth_chip_area`	Main final ASIC area target
`openroad_critical_path_delay`	Main final timing target
`openroad_total_power`	Exploratory skewed, clock-sensitive power target

The repository also extracts physical-effect descriptors from OpenROAD outputs. These include WNS, TNS, worst slack, setup and hold violation counts, maximum slew and maximum capacitance violation counts, approximate routed wirelength from final DEF files, route-guide indicators, and route-DRC report indicators.

These descriptors are not used as the main early-prediction inputs because they are post-routing quantities. Including them as model inputs would leak backend information into a predictor whose purpose is to operate before final OpenROAD implementation. Instead, they are reserved for auxiliary analysis of final physical-design behavior and its relationship to QoR.

5 Regression Modeling and Validation

The models predict continuous final QoR values, including area, delay, and power, rather than class labels. Therefore, the task is formulated as a regression problem. The main reported metric is mean absolute error (MAE), together with normalized MAE:

$$\mathrm{NMAE} = \frac{\mathrm{MAE}}{\mathrm{mean}(|y|)}. \tag{1}$$

Normalized MAE expresses the average absolute prediction error relative to the target scale. In this report, it is shown as a percentage and interpreted as normalized prediction error, not as percentage accuracy.

Several validation settings were explored, including random splits, ordinary cross-validation, and benchmark-grouped validation. The most relevant setting for the final QoR task is GroupKFold by benchmark. In this setup, each validation fold holds out complete benchmark kernels: all rows belonging to the held-out benchmark are evaluated in the test fold, while rows from that benchmark are excluded from training. This tests whether the model can generalize to benchmark kernels that were not seen during training.

Benchmark-grouped validation is more demanding than random splitting. In the experiments, random splits gave overly optimistic results because training and test rows could share benchmark-specific structure. GroupKFold reduces this similarity leakage and better represents the intended use case of predicting QoR for unseen kernels. Because the OpenROAD-labeled subset contains only 18 rows, fold sizes are small; therefore, MAE and normalized MAE are emphasized as the most interpretable error metrics.

The evaluated model families include Linear Regression, Random Forest, Gradient Boosting, Ridge Regression, Elastic Net, Gaussian Process Regression, PLS Regression, feature-selection variants, PCA variants, log-transformed input variants, and feature-group comparison experiments. The report focuses on the best direct-target group-aware results because they most directly answer the early-prediction question.

Table 5: Model families evaluated in the study

Model family	Purpose
Linear Regression	Simple baseline regression model
Ridge / Elastic Net	Regularized linear models for small datasets
Random Forest	Nonlinear tree-based baseline
Gradient Boosting	Nonlinear boosted-tree model
Gaussian Process	Small-data nonlinear regression model
PLS Regression	Regression with latent components
PCA variants	Dimensionality-reduced feature experiments
Feature-group models	Static, HLS, Yosys, ratio, and combined-feature comparisons

The project also addresses the feature-contribution objective through feature-group comparison experiments. These experiments compare models trained with different groups of intermediate features, including static-code features, Bambu/HLS metrics, Yosys synthesis statistics, HLS+Yosys features, derived ratios, and all valid pre-OpenROAD features. This analysis shows which intermediate stages provide useful predictive signal for final QoR.

The early-stage prediction objective is addressed by using only pre-OpenROAD inputs, including static source-code features, Bambu/HLS metrics, Yosys statistics, configuration variables, and derived ratios. These inputs are evaluated against final OpenROAD QoR labels, which serve only as supervised learning targets. Post-routing physical descriptors, such as routed wirelength, route-guide indicators, and DRC indicators, are not used as model inputs because they would reveal backend information. They are analyzed separately to understand how final physical-design behavior relates to QoR.

The workflow also supports the goal of backend-effort reduction. Although wall-clock speed-up was not measured directly, the surrogate model is designed to reduce the number of configurations that require full backend implementation by ranking, filtering, or prioritizing candidates before OpenROAD execution.

6 Experimental Results

This section reports the final OpenROAD QoR prediction results and interprets the relative difficulty of the area, delay, and power targets. Table 6 first summarizes the scale and distribution of the three OpenROAD targets.

Table 6: OpenROAD Target Distributions

Target	Mean	Med.	Min	Max
Area	42523.34	44335.68	11447.84	75947.79
Delay (ns)	4.4077	5.2585	2.0771	5.8108
Power (W)	0.0294	0.0057	0.00219	0.295

As shown in Table 6, the power target is visibly skewed: the mean is much larger than the median because one aggressive-clock condition produces a much larger power value.

The main direct-target results under benchmark-grouped validation are shown in Table 7. For each OpenROAD target, the table reports the model and feature setup that achieved the lowest GroupKFold error among the evaluated configurations. These are the headline feasibility results because they test whether pre-OpenROAD information can generalize to benchmark kernels that were not seen during training.

In the best-setup column, raw denotes the processed pre-OpenROAD feature matrix, ratios denotes derived-ratio features, and Yosys-only denotes models trained only with Yosys synthesis features.

Table 7: Best direct-target GroupKFold results

Target	Best setup	MAE	NMAE
Area	Elastic Net, raw	6637.11	15.34%
Delay	Elastic Net, ratios	0.763 ns	18.71%
Power	Elastic Net, Yosys-only	0.0157 W	37.70%

Figure 1 visualizes the normalized MAE values from Table 7, making the relative prediction difficulty of the three targets easier to compare.

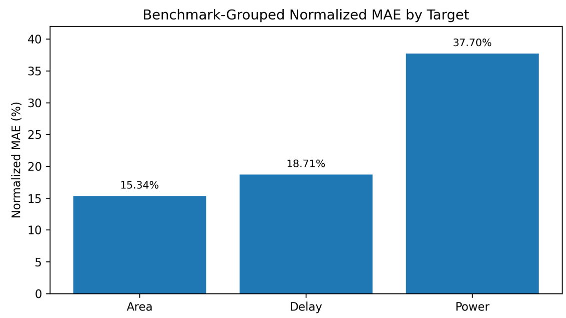

Figure 1: Normalized MAE for the three final OpenROAD QoR targets under benchmark-grouped validation

Final ASIC area is the strongest target in the current dataset. A normalized MAE of 15.34% indicates that early static, HLS, and Yosys features contain useful information about final area. This result is consistent with the fact that final area is closely related to structural properties that are already partially visible before placement and routing, such as operation count, register count, cell count, and wire-bit count.

Critical-path delay is also informative, but its error is higher than area, at 18.71% normalized MAE. Timing depends on structural depth, control/dataflow organization, synthesis mapping, and backend physical effects. Some of these signals are present in HLS and Yosys features, but final delay also depends on placement, routing, buffering, and clock constraints. Therefore, the timing result is promising but still more limited than the area result.

Total power has the largest normalized MAE at 37.70%. This should be interpreted as exploratory rather than as the main conclusion of the method. The target distribution is skewed, and the small dataset contains limited clock-condition diversity plus one aggressive-clock run. As a result, power is harder to model reliably from the current OpenROAD-labeled subset. Power should be modeled separately with transformations, robust regression, clock-normalized targets, or separate clock-regime models once more OpenROAD-labeled data is available.

7 Physical-Effect Analysis

The project also analyzes backend physical effects in addition to the main QoR prediction targets. All 18 OpenROAD runs had a final DEF (Design Exchange Format) file, a route guide, and a route DRC (design-rule checking) report. These outputs make it possible to inspect routing and DRC descriptors after implementation. Because these quantities are extracted from OpenROAD outputs, they are not used as primary early-prediction features. They are kept as auxiliary physical-effect descriptors.

Table 8 summarizes timing and electrical violation counts. No setup, hold, or max-slew violations were found in the parsed runs. Max capacitance violations were present but limited: the total count was 3, with a maximum of 2 per design.

Table 8: Timing and electrical violations

Metric	Total	Max
Setup	0	0
Hold	0	0
Max slew	0	0
Max cap.	3	2

The route DRC parser records conservative report indicators because exact OpenROAD DRC report formatting can vary. In this context, conservative means that the parser does not assume a detailed violation count unless clear violation-related keywords are present. Twelve designs were classified as clean or empty DRC reports, and six were classified as possible DRC violations. Across the 18 designs, there were 7 DRC keyword hits, with at most 2 hits in a single design.

Table 9: Route artifact and DRC summary

Metric	Value
Final DEF files	18
Route guides	18
DRC reports	18
Clean / empty DRC reports	12
Possible DRC violations	6
DRC keyword hits	7
Max hits/design	2

Approximate routed wirelength is computed from coordinate sequences in the final DEF files by summing Manhattan distances between consecutive coordinate pairs. This is not an official router-reported wirelength value, so it is treated as an approximation. However, it provides a consistent routing-complexity indicator across the parsed OpenROAD runs. The wirelength distribution is summarized in Table 10.

Table 10: Approximate routed wirelength

Statistic	Microns
Mean	35498445.9592
Median	35520496.4200
Min	32631324.1450
Max	38785191.2250

Selected high physical-effect correlations with final QoR are shown in Table 11 and Figure 2. Routing-complexity indicators are strongly correlated with final area and moderately correlated with critical-path delay.

Table 11: Selected Physical-Effect Correlations with Final QoR

Metric	QoR	Corr.
Routed nets	Area	0.9978
Guide nonempty	Area	0.9974
Guide segments	Area	0.9968
Wirelength	Area	0.9315
DEF pairs	Delay	0.7149
Guide segments	Delay	0.7083

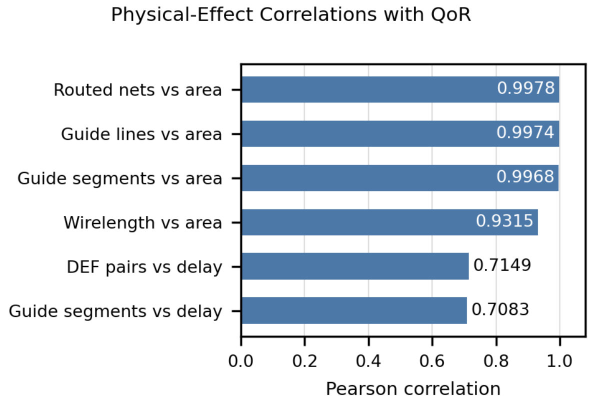

Figure 2: Selected Pearson correlations between post-routing physical descriptors and final QoR metrics

These correlations show that routing complexity and final physical design structure are strongly related to final area, with a moderate relationship to final timing. However, because routed wirelength, route-guide lines, and DRC indicators are only available after OpenROAD execution, they should not be used as main inputs for an early surrogate model. A possible improvement is to develop pre-routing proxies for these physical effects so that routing complexity can be estimated before final routing.

8 Discussion and Limitations

The main result is that pre-OpenROAD information carries useful signal for final area and timing. Area is easier to predict because it is strongly related to structural features that are visible in HLS and synthesis reports. Timing is harder because critical-path delay depends not only on logic structure but also on placement, routing, buffering, and physical constraints. The area result is especially promising because it achieves 15.34% normalized MAE under benchmark-grouped validation, despite the small OpenROAD-labeled subset.

The most important limitation is the small number of OpenROAD-labeled runs. The 18 backend samples are enough for an initial feasibility study, but not enough for a production-level predictor.

This affects both model training and validation stability, especially for power. It also means that the reported errors may be sensitive to the exact benchmark grouping and to the inclusion of the aggressive-clock power outlier.

Another limitation is the lack of direct runtime measurements. The workflow motivates backend-run reduction, but the report does not directly measure wall-clock speed-up from surrogate-guided candidate selection. A more complete speed-up study would require measuring full backend runtimes and comparing them with a surrogate-assisted candidate-selection workflow under the same compute environment.

The current static source-code features are extracted using lightweight pattern-based parsing. This choice was practical for a feasibility study because it allowed consistent feature extraction across the PolyBench/C kernels without adding a separate compiler-analysis infrastructure. More advanced AST-based or LLVM-based analysis could be added in future work to capture program structure more precisely, such as loop nests, memory-access patterns, data reuse, induction variables, and data dependencies.

9 Recommendations for Reducing Error

The most important way to reduce error is to increase the number of OpenROAD-labeled backend runs. The current OpenROAD-labeled subset is small and mainly covers the benchmark-diverse 10 ns backend condition, with one additional aggressive-clock run. More backend samples would broaden the training distribution, reduce fold-to-fold sensitivity in GroupKFold validation, and let the models learn configuration effects instead of relying heavily on benchmark identity. The expanded subset should be selected systematically across clock periods, memory policies, optimization levels, dataset sizes, DSP coefficients, and backend constraints.

Backend runtime can also be used more efficiently with active learning or uncertainty sampling. After an initial model is trained, new OpenROAD runs can prioritize configurations where predictions are uncertain, where model families disagree, or where the predicted design point is close to an area-delay Pareto frontier. This would increase the information gained from each expensive backend run.

Feature quality can also be improved in future iterations. The current static feature extractor uses lightweight source-pattern counts, which was practical for building the first complete HLS-to-OpenROAD modeling workflow. A more advanced version could move toward AST- or LLVM-based features that capture loop nests, memory-access patterns, data reuse, induction variables, and operation structure more accurately. Yosys features could also be enriched with structural timing proxies such as logic depth, fanout, connectivity, memory/register organization, and critical combinational path estimates.

Finally, future models should include pre-routing proxies for routing complexity and should treat power separately. Netlist connectivity, hierarchy, fanout, memory structure, and placement-independent graph features may approximate the routed-wirelength and route-guide effects observed after OpenROAD. Power should be modeled with log transformations, robust regression, clock-normalized targets, or separate clock-regime models because the current raw power distribution is skewed and clock-sensitive.

10 Conclusion

This project built a complete source-to-backend HLS-to-ASIC QoR surrogate modeling workflow. It generates and parses Bambu/HLS configurations, extracts static C-code features, parses Yosys synthesis reports, prepares a benchmark-diverse subset for OpenROAD backend execution, parses final OpenROAD QoR reports, merges and cleans the datasets, trains regression models, and analyzes post-routing physical effects. The result is a complete experimental pipeline for evaluating whether early HLS, static, and synthesis information can predict final ASIC/OpenROAD outcomes.

The results show useful predictive signal for final area and timing. Under GroupKFold validation by benchmark, final ASIC area achieved 6637.11 MAE, corresponding to 15.34% normalized MAE. Critical-path delay achieved 0.763 ns MAE, corresponding to 18.71% normalized MAE. Power remains exploratory, with 0.0157 W MAE and 37.70% normalized MAE, because the observed power distribution is skewed and sensitive to clock conditions.

The physical-effect analysis showed that routing-complexity indicators are strongly correlated with final area and moderately correlated with final timing. Routed net count, route-guide indicators, and approximate routed wirelength all show that final physical design behavior is important for QoR. These metrics are post-routing descriptors, so they are not used as main early inputs, but they motivate pre-routing proxies for routing complexity.

The main limitation is the small OpenROAD-labeled subset. Therefore, the current study should not be interpreted as a mature predictor for production use. However, the achieved area and critical-path delay errors are meaningful for an initial regression study, especially because they were obtained under benchmark-grouped validation where complete benchmark kernels are held out from training. This shows that the project not only builds the data pipeline, but also demonstrates practical early-prediction potential for final ASIC area and timing.

References

[1] Panda Project, *Bambu PolyBench results*. Available: https://panda.deib.polimi.it/?page_id=1152

[2] Panda Project, *Bambu: A free framework for the high-level synthesis of complex applications*. Available: https://panda.deib.polimi.it/?page_id=31

[3] YosysHQ, *Yosys Open SYnthesis Suite*. Available: https://yosyshq.net/yosys/

[4] The OpenROAD Project, *OpenROAD*. Available: https://theopenroadproject.org/

[5] The OpenROAD Project, *OpenROAD-flow-scripts*. Available: https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts

[6] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.